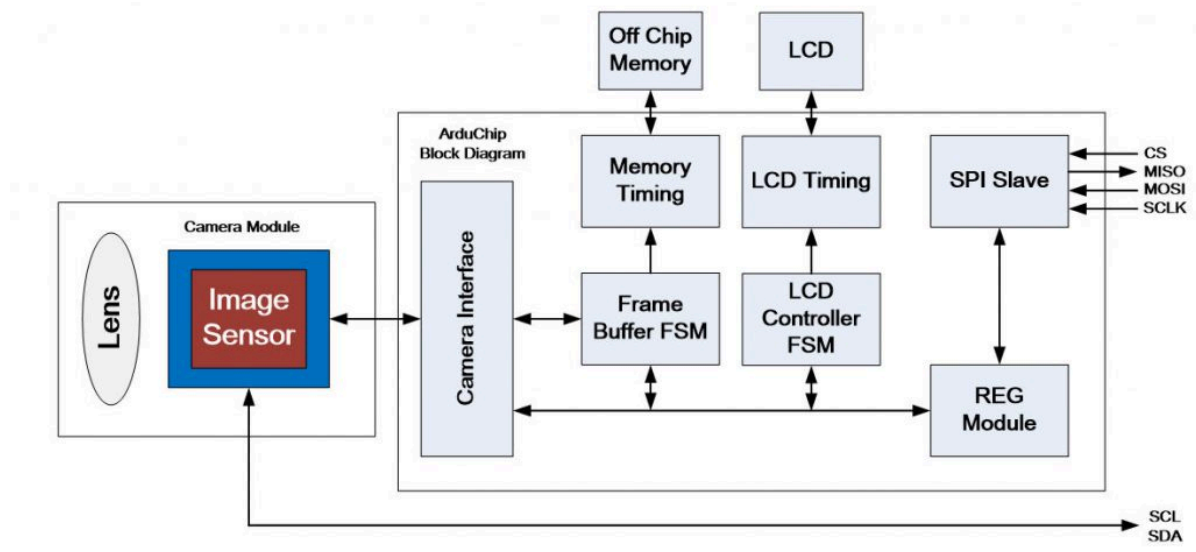


Arducam Chip

1. Introduction

The ArduChip is ArduCAM proprietary camera controller technology which hides the complex camera, LCD screen, memory and user interface hardware timing into a chip level and provides a set of control registers and user friendly SPI interface.



2. SPI Interface Timing

The ArduCAM SPI slave interface is fixed SPI mode 0 with POL = 0 and PHA = 0. The maximum speed of SCLK is designed for 8MHz, care should be taken do not over clock the maximum 8MHz. The SPI protocol is designed with a command phase with variable data phase. The chip select signal should always keep asserted during the SPI read or write bus cycle. The first bit[7] of the command phase is read/write byte, '0' is for read and '1' is for write, and the bit[6:0] is the address to be read or write in the data phase.

2.1 SPI Bus Write Timing

The SPI bus write timing composed of a command phase and a data phase during the assertion of the chip select signal CSn. The first 8 bits is command byte which is decoded as a register address, and the second 8 bits is data byte to be written to the ArduChip internal registers.

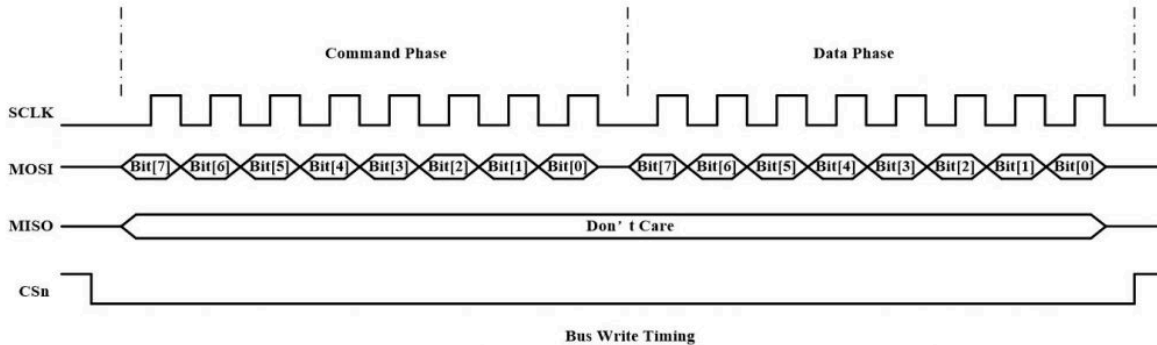


Figure 1 SPI Bus Write Timing

2.2 SPI Bus Single Read Timing

The SPI bus single read timing is for read operation of ArduChip internal registers and single FIFO read function. It is composed of a command phase and a data phase during the assertion of chip select signal CSn. The first 8 bits is command byte which is decoded as a register address, the second 8 bits is dummy byte written to the SPI bus MOSI signal, and the content read back from register is appeared on the SPI bus MISO signal.

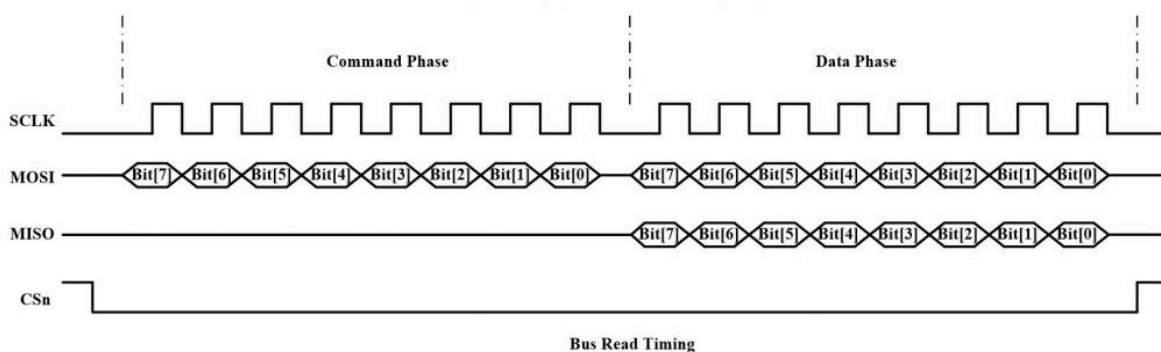


Figure 2 SPI Bus Single Read Timing

2.3 SPI Bus Burst Read Timing

The SPI bus burst read timing is only apply for burst FIFO read operation. It is composed of a burst read command phase and multiple data phases in order to get double throughput compared to the single FIFO read operation. (Note: Older Arducam spi cameras, the first byte read from the FIFO is a dummy byte, and the following bytes are valid bytes.)

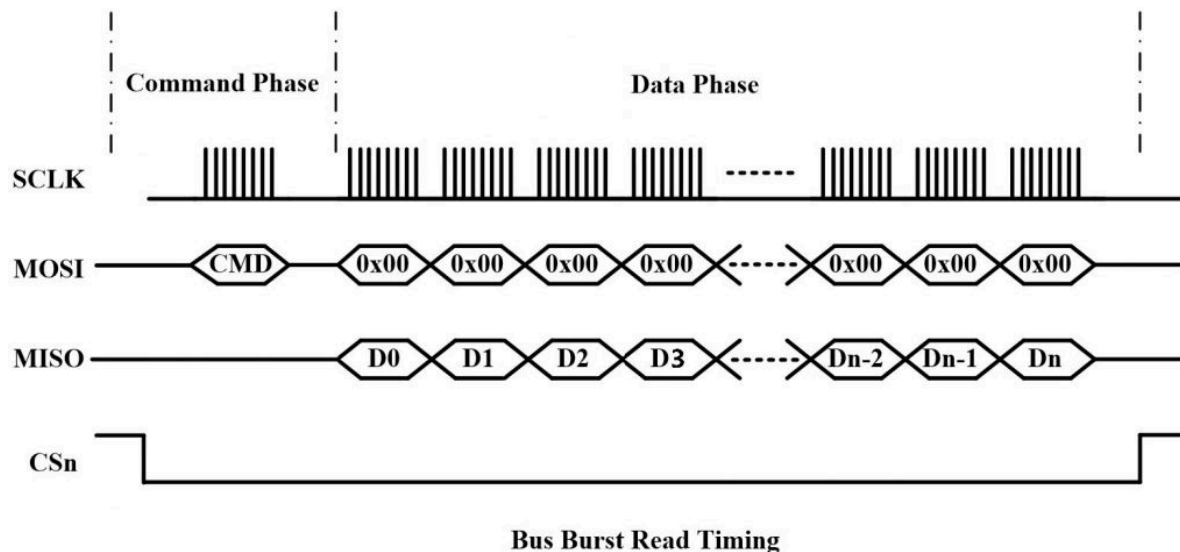


Figure 3 SPI Bus Burst Read Timing

3. Registers Table

Sensor and FIFO timing is controlled by a set of registers which is implemented in the ArduChip. User can send capture commands and read image data with a simple SPI slave interface. The detail description of registers' bits can be in this section.

As mentioned earlier the first bit[7] of the command phase is read/write byte, '0' is for read and '1' is for write, and the bit[6:0] is the address to be read or write in the data phase. So user has to combine the 8 bits address according to the read or write commands they want to issue.

Different ArcuCAM platform uses different ArduChip and has different functionalities. Here is a list of possible hardware platforms:

Hardware Platform	Functions				
Single Capture/	Burst Read	Multiple Capture	Rewind	Low Power Mode	Short Video

Hardware Platform	Functions					
	Read					Capture
ArduCAM Shield Rev.C	√					
ArduCAM Shield Rev.C+	√	√	√	√	√	
ArduCAM- Mini-2MP	√	√			√	√
ArduCAM- Mini-5MP	√	√	√	√	√	√
ArduCAM- Mini-5MP (Bit- Rotation- Fixed)	√	√	√	√	√	√
ArduCAM Shield V2	√	√	√	√		
ArduCAM- Mini-5MP- Plus (OV5642)	√	√	√	√	√	√
ArduCAM- Mini-5MP- Plus (OV5640)	√	√	√	√	√	√

It is a basic capture function of the ArduChip. The capture command code is 0x84, and write '1' to bit[1] to start a capture sequence. And then polling bit[3] which is the capture done flag by sending command code 0x41. After capture is done, user have to clear the capture done flag by sending command code 0x41 and write '1' into bit[0] before next capture command.

By sending the command code 0x81 and with writing the number of images to be capture into bit[2:0], before starting the capture command as the single capture sequence does. Please note that user should trade off between the resolution and number of images to be captured and do not make the frame buffer overflow.

Use the same command as the Multiple Capture Mode. When the value bit[2:0] equals to 7, the ArduCAM will continuously capture the images until the entire frame buffer is full. User can save the captured MJPEG to AVI files to create short movie clips.

It is basic memory read function which start a single read operation and read a single byte each time. By sending command code 0x3D to start a single read operation, a single byte is read out from the frame buffer.

It is advance capture function which can read multiple bytes out of the frame buffer by just sending a single command code 0x3C.

Please note that for these hardware platforms (ArduCAM Shield Rev.C+, ArduCAM-Mini-2MP, ArduCAM-Mini-5MP) the first read byte should be ignored in the first read transaction, because it is a dummy byte. In the following read transaction, the first byte read is the last read byte in the last read transaction, it is very important. And do not use other SPI command between burst read transaction. Detail timing can be found from Figure 5.

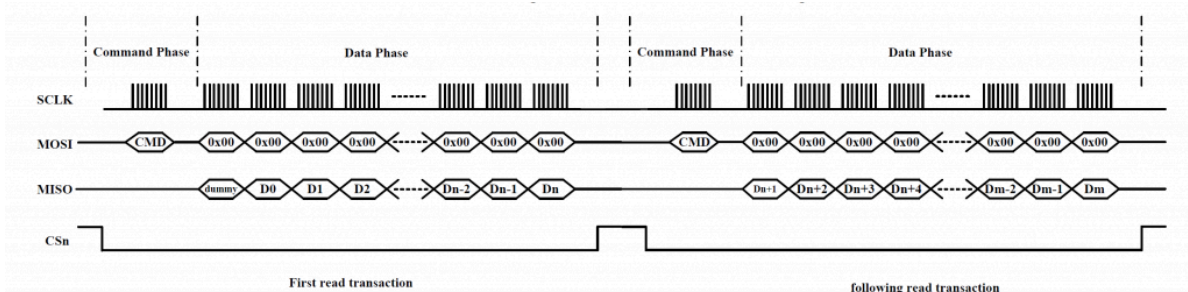


Figure 5 Burst read timing diagram 1

For hardware platforms (ArduCAM-Shield-V2, ArduCAM-Mini-5MP-Plus), you don't need to worry about the first byte. Detail timing can be found from Figure 6.

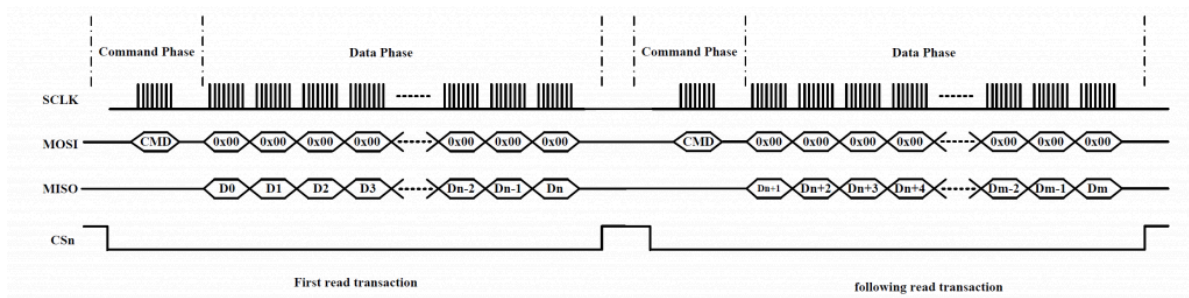


Figure 6 Burst read timing diagram 2

Rewind read is useful for some application that need access the same pixel data multiple times. By sending the command code 0x84 and write '1' to bit[5] in the data phase, it will reset the memory read pointer to ZERO. Then user can read the image data from the start of the memory.

For some battery powered device power consumption is very important. There are two levels to achieve low power mode, user have to combine these modes according to their own power strategy.

It is achieved by controlling the power enable pin of the onboard LDOs. The power enable pin is controlled by the GPIO[2] of ArduChip. By sending the command code 0x86 and write '1' to bit[2] to enable the LDOs, or write '0' to bit[2] to disable the LDOs to save power. Note that power down the sensor circuit, the camera settings are lost. User should reinitialize the sensor when power up the sensor circuit again.

It is achieved by controlling the power enable pin of the onboard LDOs. The power enable pin is controlled by the GPIO[1] of ArduChip. By sending the command code 0x86 and write '1' to bit[1] to set the sensor into standby mode, or write '0' to bit[1] to set the sensor out of standby mode. Note that the sensor settings are not lost when in standby mode, and reinitialize is not needed.

Register Table

Sensor and FIFO timing is controlled with a set of registers which is implemented in the ArduChip. User can send capture commands and read image data with a simple SPI slave interface. The detail description of registers' bits can be found in the software section in this document. Not all the registers are implemented in a given hardware platform, please check the hardware develop guide for detail register description for certain hardware you've got. As mentioned earlier the first bit[7] of the command phase is read/write byte, '0' is for read and '1' is for write, and the bit[6:0] is the address to be read or write in the data phase. So

user has to combine the 8 bits address according to the read or write commands they want to issue.

Table 1 ArduChip Register Table

Register Address	Register Type	Mini_2mp	Mini_2mp_plus	Mini_5mp_plus	Shield V2
0x00	RW	Test Register			
0x01	RW	/	Capture Control Register Bit[7:0]: Number of frames to be captured The value in this register + 1 equal to the number of frames to be captured. The value=255 means capture continuous frames until the frame buffer is full, it is used for short video clip recording. (Mini_5mp_plus_Altera: value=7)		
0x02	RW	/	/	Sensor Interface Timing Register Bit[0]: Sensor Hsync Polarity, 0 = active high, 1 = active low Bit[1]: Sensor Vsync Polarity 0 = active high, 1 = active low Bit[3]: Sensor PCLK reverse 0 = normal, 1= reversed PCLK	
0x03	RW	/	Sensor Interface Timing Register Bit[0]: Sensor Hsync Polarity, 0 = active high, 1 = active low Bit[1]: Sensor Vsync Polarity 0 = active high, 1 = active low Bit[3]: Sensor PCLK reverse 0 = normal, 1= reversed PCLK		Sensor Interface Timing Register Bit[0]: Sensor Hsync Polarity, 0 = active high, 1 = active low Bit[1]: Sensor Vsync Polarity 0 = active high, 1 = active low Bit[2]: LCD backlight enable 0 = enable, 1 = disable Bit[3]: Sensor PCLK reverse 0 = normal, 1= reversed PCLK
0x04	RW	FIFO control Register			

		Bit[0]: write '1' to clear FIFO write done flag Bit[1]: write '1' to start capture Bit[4]: write '1' to reset FIFO read pointer			
0x05	RW	/	Test Mode Register Bit[0]: 0 = CAM, 1 = Test Data		
0x06	RW	GPIO Write Register Bit[0]: Sensor reset IO value Bit[1]: Sensor power down IO value Bit[2]: Sensor power enable IO value			/
0x07	RW	/	Bit[7]: write '1' to reset CPLD		
0x08	RW	/			LCD high 8bits
0x3C	RO	Burst FIFO read operation			
0x3D	RO	Single FIFO read operation			
0x3E	WO	/	/	/	LCD control register with RS=0 LCD control register with RS=1
0x3F	WO	/	/	/	
0x40	RO	ArduChip version Bit[7:4]: integer part of the revision number Bit[3:0]: decimal part of the revision number Mini_2mp(UC-272 Rev.B), 0x55 Mini_2mp_plus(UC-474 Rev.A), 0x73 Mini_5mp_plus(UC-367 Rev.E), 0x73			
0x41	RO	Bit[0]: camera vsync pin status Bit[3]: camera write FIFO done flag	Bit[0]: camera vsync pin status Bit[1]: external trigger status Bit[3]: camera write FIFO done flag		
0x42	RO	Camera write FIFO size[7:0]			
0x43	RO	Camera write FIFO size[15:8]			
0x44	RO	Camera write FIFO size[18:16]	Camera write FIFO size[23:16]		
0x45	RO	/	FIFO Status Register Bit[0]: FIFO full status 0 = unfull, 1 = full		
0x46	RO	/	ArduChip version year Bit[6:0]: year - 2000		
0x47	RO	/	ArduChip version month Bit[3:0]: month		
0x48	RO	/	ArduChip version date		

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